THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 6: **Control and Coordination** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Reflex actions are not controlled by the brain directly. Reason (R): The reflex arc connects the receptor to the effector through the spinal cord. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A hormone X is secreted by an organ below the stomach, regulates sugar. X's deficiency causes: 1

(a) dwarfism **(b)** goitre **(c)** diabetes **(d)** gigantism

3. A boy has a rapid pulse, sweating and wide pupils when frightened. This is due to: 1

(a) insulin **(b)** thyroxine **(c)** oestrogen **(d)** adrenaline

4. Assertion (A): Nerve impulse travels from dendrite to axon end in a neuron. Reason (R): Chemical signals cross the synapse in only one direction. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. A student touches a hot object, withdraws the hand and then feels the heat. The reason is: 1

(a) motor neuron is slower
(b) brain acts before spinal cord
(c) no relay neuron
(d) reflex action is faster than brain sensation

6. A woman has swollen neck and slow metabolism. Her diet is deficient in: 1

(a) calcium **(b)** iron **(c)** zinc **(d)** iodine

7. Assertion (A): Plant roots show positive geotropism. Reason (R): Roots grow upwards against gravity. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. An impulse is blocked in the motor neuron only. The patient will: 1

(a) feel more **(b)** feel the touch but not move the muscle
(c) neither feel nor move **(d)** move but not feel

9. **Assertion (A):** A reflex action is a quick, automatic response. 1
Reason (R): In a reflex action the signal passes through the spinal cord and not through the brain.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Auxin is synthesised at the shoot tip. 1
Reason (R): Gibberellins help in the growth of the stem.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A patient cannot pull his hand away from a hot object but can feel the heat. Which part of the reflex arc is damaged? 2

12. Why is the hormone from a gland not released continuously in the same amount? 2

13. Why do we sweat and breathe fast when frightened? 2

14. Why does the roots' growth against light (negative phototropism) benefit the plant? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A seedling is kept in a box with a window on one side; after a few days it bends. When the tip was cut, no bending was observed. Explain. 3

16. A frog's leg muscle twitches when a nerve is stimulated. Explain which nerve and the reflex path. 3

17. Differentiate between the effects of adrenaline and insulin on blood sugar. 3

OR

Explain how the nervous system and hormones act together when we see a snake. 3

18. How do plants ensure that the pollen tube reaches the ovule? 3

OR

Why is the sense of taste affected when you have a cold? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** While helping in the kitchen, Ramesh accidentally touched a hot tava and immediately pulled his hand away, even before he realised that it was hot. 5
(i) Name this type of response.
(ii) Name the neuron that carries the impulse from the spinal cord to the muscle of the hand.
(iii) Which part of the nervous system acts as the main centre for this action?
(iv) State two differences between this action and writing a letter.

20. A person touches a hot plate, quickly pulls the hand and then screams. Explain the sequence in terms of reflex arc and brain involvement with a flow chart. 5

OR

Explain how the feedback mechanism controls hormone release with examples of insulin and thyroxine. 5

END OF QUESTION PAPER — Total: 40 Marks